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Engineering resistance by simultaneous editing of two susceptibility gene promoters against bacterial leaf streak in rice

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Xanthomonas oryzae pv. *oryzicola* (Xoc) uses a type III secretion system to inject virulent proteins, named transcription activator-like effectors (TALEs), into plant cells that transcriptionally activate susceptibility (S) genes helping pathogen proliferation, causing bacterial leaf streak (BLS) of rice. Successful induction of the S gene requires binding of TALE(s) to the effector binding element (EBE) sequence in the gene promoter, then recruiting host transcriptional machinery for gene activation. This transcriptional reprogramming generates a series of molecular processes, providing a cascade of nutrition to bacteria, or may diminish the plant resistance layer by minimizing defense genes. Resistance resources against Xoc are scarce and largely unexplored, seeking pathogen attacks and conferring significant yield losses. Genome editing provides new horizons for engineering resistance against bacterial diseases of rice. CRISPR/Cas9-mediated genome used to simultaneously edit EBEs of two important S genes for Xoc, one belonging to sulfate transporter gene family OsSULTR3;6 and another encoding 2-oxoglutarate-dependent dioxygenase OsF3H03g, in susceptible rice cultivar Nipponbare (Nip). Three double mutant transgene-free lines named Δ DMT2-13, Δ DMT2-15, and Δ DMT220 were obtained, having deletions of 1-13, 24-8, and 24-4 bp in EBE of both genes. All these three mutant plants confer robust resistance against a multitude of Xoc strains producing significantly smaller lesion length as compared to Nip. Quantitative real-time PCR of $\Delta\Delta$ EBE plants informed nonsignificant gene induction after Xoc inoculation, while Nip showed 40-to-60-fold increased transcripts levels of genes at 24hpi. Significantly higher bacterial colony-forming units(cfu) were observed in Nip, while mutants showed lower cfu at 2, 4, 6, 8, and 14th day of inoculation. To ascertain that the OsSULTR3;6 induction provides susceptibility, we developed designer-TALE-SUL3;6 (dTALE) for its artificial induction. Transforming dTALE-SUL3;6 into RS105 and inoculation produced increased lesion length as compared to RS105 in mutant plants. Moreover, in field conditions, mutant plants yield and agronomic traits were not affected, implying that $\Delta\Delta$ EBE causes no significant fitness cost.